

Bike Availability & Station Analysis

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Project Overview

The **Bike Availability & Station Analysis Dashboard** is an interactive Business Intelligence solution developed using **Power BI** to analyze the performance of public bike-sharing stations across multiple cities. The dashboard provides real-time insights into bike availability, station capacity, operational status, and geographic distribution, enabling stakeholders to optimize bike allocation and improve operational efficiency.

Project Objectives

- Monitor bike availability across all stations.
 - Analyze station operational status (Open vs Closed).
 - Compare available bikes with total bike stand capacity.
 - Identify top-performing and low-performing cities.
 - Visualize bike station locations using an interactive map.
 - Support data-driven decision-making for bike-sharing operations.
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Tools & Technologies

- Microsoft Power BI
 - Power Query
 - DAX (Data Analysis Expressions)
 - Data Visualization
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Dataset Information

The dataset contains detailed information about bike-sharing stations across multiple cities.

Dataset Columns

- Station Number
- Station Name
- Contract Name (City)
- Address
- Latitude
- Longitude
- Bike Stands
- Available Bikes
- Available Bike Stands
- Station Status
- Bonus

Data Cleaning & Transformation

The dataset was cleaned and transformed using **Power Query**.

Data Preparation Steps

- Removed duplicate records.
 - Handled missing and null values.
 - Corrected data types.
 - Standardized column names.
 - Removed unnecessary columns.
 - Validated station status values.
 - Extracted Latitude & Longitude from the Position column.
 - Prepared latitude and longitude for map visualization.
 - Optimized the dataset for reporting and analysis.
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Data Modeling

A structured Power BI data model was created to improve report performance.

Measures Created

- Total Bike Stations
 - Total Bike Stands
 - Total Available Bikes
 - Total Available Bike Stands
 - Open Stations
 - Closed Stations
 - Bike Availability %
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Dashboard KPIs

KPI	Value
 Total Bike Stations	3,078
 Total Bike Stands	57,203
 Available Bikes	19,646
 Available Bike Stands	32,157
 Open Stations	2,872
 Closed Stations	206
 Bike Availability	34.34%

Dashboard Features

- KPI Cards
- Contract-wise Bike Availability Analysis
- Bike Stands vs Available Bikes Comparison

- Station Status Distribution
 - Interactive Geographic Map
 - Contract-wise Summary Table
 - Dynamic Filters and Slicers
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Key Insights

- The bike-sharing network consists of **3,078 bike stations** with a total capacity of **57,203 bike stands**.
 - Currently, **19,646 bikes** are available for users across all monitored stations.
 - There are **32,157 available bike stands**, indicating sufficient docking capacity for returning bikes.
 - **2,872 stations (93.31%)** are operational, while **206 stations (6.69%)** are currently closed.
 - The overall **Bike Availability Rate is 34.34%**, meaning approximately one-third of the total docking capacity currently has bikes available.
 - **Bruxelles-Capitale** records the highest number of available bikes and bike stands, making it the best-performing contract location.
 - **Lyon** and **Dublin** also maintain high bike availability, reflecting strong service coverage and operational efficiency.
 - Most stations (**99.44%**) are classified as **Non-Bonus Stations**, while only **0.56%** are Bonus Stations.
 - The interactive map highlights bike station distribution across multiple European cities, supporting regional operational analysis.
 - The dashboard enables quick identification of cities with low bike availability, helping optimize bike redistribution strategies.
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Business Analysis

- The dashboard provides a comprehensive overview of station capacity, available bikes, operational status, and city-wise performance through interactive visualizations and KPI cards.
- The network consists of **3,078 bike stations** with a total capacity of **57,203 bike stands**.
- Currently, **19,646 bikes** are available for users.
- There are **32,157 available bike stands**, indicating vacant docking capacity.
- **2,872 stations (93.31%)** are operational, while **206 stations (6.69%)** are closed.

- The overall **Bike Availability Rate is 34.34%**, showing that approximately one-third of the total docking capacity currently contains available bikes.
 - The analysis reveals differences in bike availability across cities, identifies closed stations, and compares available bikes with docking capacity to understand station utilization.
 - Historical station availability trends can be leveraged to forecast future bike demand, optimize redistribution schedules, and improve service reliability during peak periods.
 - **Bruxelles-Capitale** has the highest station capacity with **8,592 bike stands** and **3,532 available bikes**, indicating the largest bike-sharing infrastructure.
 - **Dublin** maintains **3,668 bike stands** with **1,241 available bikes**, demonstrating strong operational performance.
 - The presence of **206 closed stations** reduces the overall availability of bikes and service coverage.
 - Approximately **56.22% of bike stands are vacant (32,157 out of 57,203)**, suggesting opportunities to rebalance bike distribution between stations.
 - Nearly **99.44% of stations are Non-Bonus Stations**, while only **0.56%** are Bonus Stations.
 - With only **34.34% bike availability**, high-demand stations may experience bike shortages during peak hours.
 - Cities with higher station capacity, such as **Bruxelles-Capitale** and **Dublin**, are expected to continue handling the highest bike demand.
 - If the **206 closed stations** remain unavailable, operational efficiency and customer accessibility may decline.
 - Regular monitoring of bike availability can help forecast demand and improve redistribution planning.
 - Increase bike allocation in high-demand cities such as **Bruxelles-Capitale** and **Dublin**. Expand bonus stations in high-traffic areas.
 - Reopen or investigate the **206 closed stations** to improve service availability.
 - Balance bike distribution across cities to reduce shortages.
 - Continuously monitor station utilization to improve operational efficiency.
 - Redistribute bikes from stations with excess capacity to locations with low bike availability.
 - Expand the number of **Bonus Stations**, currently only **0.56%**, to encourage better bike circulation.
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Business Benefits

- Improves visibility into bike-sharing operations.
 - Supports efficient bike redistribution.
 - Enhances station performance monitoring.
 - Helps reduce bike shortages.
 - Enables data-driven operational planning.
 - Improves customer satisfaction through better bike availability.
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Future Enhancements

- Integrate real-time bike availability APIs.
 - Develop demand forecasting using machine learning.
 - Add hourly and daily trend analysis.
 - Include weather impact on bike usage.
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Conclusion

The **Bike Availability & Station Analysis Dashboard** transforms raw operational data into meaningful business insights through interactive Power BI visualizations. By analyzing **3,078 stations**, **57,203 bike stands**, and **19,646 available bikes**, the dashboard provides a clear understanding of station utilization, operational status, and bike distribution. With a **34.34% bike availability rate** and **93.31% operational stations**, the solution supports informed decision-making, efficient resource allocation, and improved bike-sharing services. It demonstrates practical skills in **Power BI, Power Query, DAX, data modeling, and business intelligence**, making it a strong portfolio project for data analytics roles.
